ForgeDB

Raw Data to PostgreSQL — Automated Schema Generation

Angular | ASP.NET Core | Python FastAPI | PostgreSQL

MVP Development Plan

3-Month Roadmap | 12 Weeks | 10 Core Features

Project Overview

ForgeDB is a system that helps users transform raw data from Excel/CSV files or APIs into a real, organized PostgreSQL database. The system analyzes the data, suggests tables, columns, data types, constraints, and relationships — then presents these suggestions to the user for review and editing before generating the actual database.

> ### MVP Goal
>
> Build a fully working first version that proves the core concept:
> **Upload Data → Analyze → Suggest Schema → User Review → Generate PostgreSQL Database**

System Architecture

Layer	Technology	Role
Angular Frontend	Angular + TypeScript + Tailwind CSS	User interface & interactions
Backend API	ASP.NET Core Web API + EF Core + JWT	Business logic, auth, orchestratio
Analysis Service	Python FastAPI + Pandas + NumPy + Regex	Data profiling & schema suggestio
Database	PostgreSQL + Npgsql	Storage & generated databases

`Communication Flow:` `Angular → ASP.NET Core API (REST) → Python FastAPI (internal HTTP) → PostgreSQL (Npgsql SQL)`

The user uploads a file via Angular. ASP.NET Core saves it and forwards it to Python FastAPI for analysis. Python returns profiling results and schema suggestions. ASP.NET stores results in PostgreSQL and returns them to the frontend. After user approval, ASP.NET generates and executes the SQL directly on PostgreSQL.

Core Features

01 User Authentication

- Register / Login with JWT
- Manage personal projects
- ASP.NET Core Identity + PostgreSQL

02 Project Management

- Create and manage multiple projects
- Track: files, analysis results, schema, status
- Entity Framework Core + PostgreSQL

03 Data Import

- Upload CSV and Excel files
- Import from API URL returning JSON
- Angular + ASP.NET + Python (Pandas, OpenPyXL, Requests)

04 Data Profiling

- Extract column names, row count, data types
- Calculate Null ratio, Unique ratio, Duplicates
- Show data examples per column

05 Schema Suggestion

- Suggest table & column names, data types
- Detect Primary Key, NOT NULL, UNIQUE
- Rule-based engine using Pandas + Regex

06 Relationship Detection

- Match column names & values across tables
- Detect FK patterns (id, user_id, customer_id…)
- Manual relationship addition when auto-detection fails

07 Data Quality Report

- Missing values & duplicate report
- Format issues (dates, phone, email)
- Overall data quality score

08 User Review Screen

- Edit table/column names and data types
- Modify PK, FK, NOT NULL, UNIQUE
- Add relationships manually via UI

09 Database Generation

- Execute CREATE TABLE, INSERT DATA, Constraints
- Full PostgreSQL Transaction with Rollback on failure
- Dynamic SQL Generation via Npgsql

10 Dashboard

- Project status, table & row counts
- Data Quality Score with charts
- Chart.js / ApexCharts + Angular

3-Month Development Timeline

Month 1	Foundation & Data Import
Week 1	Project setup — GitHub repo, Angular, ASP.NET, PostgreSQL, Python FastAPI
Week 2	Authentication — Login/Register, JWT, User Projects API, Project UI
Week 3	File Upload API — CSV, Excel upload, Python reading, File metadata
Week 4	API Import — JSON reading, Data preview, Angular ↔ Backend connection

Month 2	Analysis & Schema Suggestion
Week 5	Data Profiling Engine — Column analysis, Data types, Null/Unique ratio, Duplicates
Week 6	Schema Suggestion Engine — Tables, Columns, Data types, NOT NULL, UNIQUE
Week 7	PK/FK Detection — Value matching, Relationship discovery, Manual relations UI
Week 8	Data Quality Report — Missing values, Duplicates, Format issues, Quality score

Month 3	Database Generation & Finalization
Week 9	Review Screen — Edit tables, columns, relations, Save final schema
Week 10	SQL Generation — CREATE TABLE, Constraints, INSERT DATA, Rollback support
Week 11	Dashboard — Project status, Quality charts, DB info, UI/UX polish
Week 12	Testing, Bug fixing, Documentation, Final presentation, GitHub README

Technology Stack

Layer	Technologies
Frontend	Angular · TypeScript · Tailwind CSS · Angular Reactive Forms · Chart.js / ApexCl
Backend	ASP.NET Core Web API · C# · Entity Framework Core · JWT Authentication · Np
Analysis Service	Python · FastAPI · Pandas · NumPy · OpenPyXL · Regex · Requests
Database	PostgreSQL
Tools	GitHub · Postman · Swagger · Docker (future)

Out of Scope (MVP)

- ✗ AI Chatbot or Machine Learning models
- ✗ Support for MySQL / SQL Server
- ✗ Big Data processing
- ✗ Real-time Streaming
- ✗ Advanced Cloud Deployment
- ✗ Multi-tenant isolation

MVP Success Criteria

1	User registers and logs in
2	User creates a new project
3	User uploads a CSV or Excel file
4	User views the data analysis results
5	User views the suggested schema
6	User edits and adjusts the suggestions
7	User clicks Generate Database
8	System creates a real PostgreSQL database
9	User views the generated tables and data